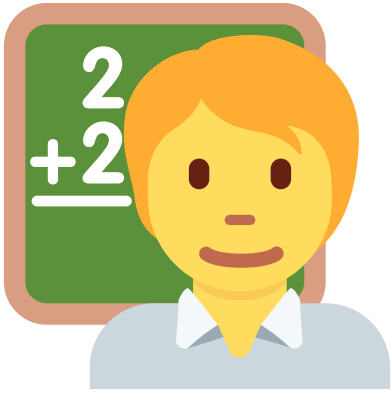


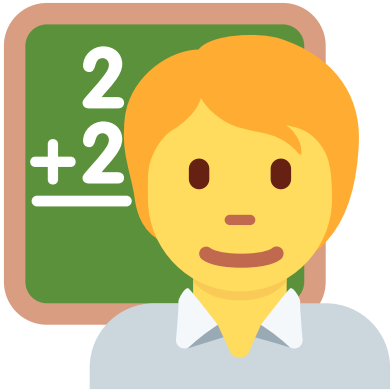
if/elif/else



Human Calculator

What is `printend`?

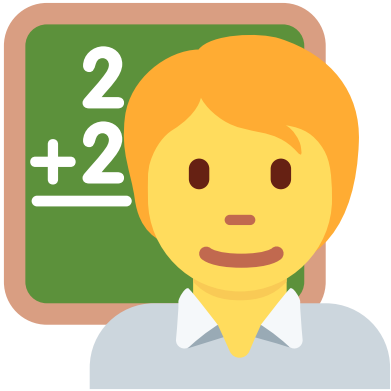
```
x = 20
if x < 20:
    msg = "A"
elif x <= 20:
    msg = "B"
else:
    msg = "C"
print(msg)
```



Human Calculator

What is printend?

```
x = 19
msg = "X"
if x < 20:
    msg = "A"
if x < 19:
    msg = "B"
else:
    msg = "C"
print(msg)
```



Human Calculator

What is printend?

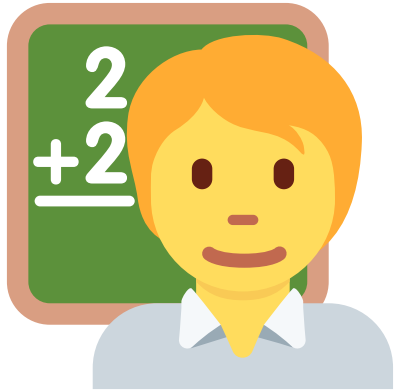
```
temp = 22
if temp > 20:
    if temp < 25:
        res = "A"
else:
    res = "B"
print(res)
```



Code Wizard

Optimize

```
temp = 22
if temp > 20:
    if temp < 25:
        res = "A"
else:
    res = "B"
print(res)
```



Human Calculator

What is printed?

```
score = 90
if score >= 80:
    grade = "A"
elif score >= 90:
    grade = "A+"
else:
    grade = "B"
print(grade)
```

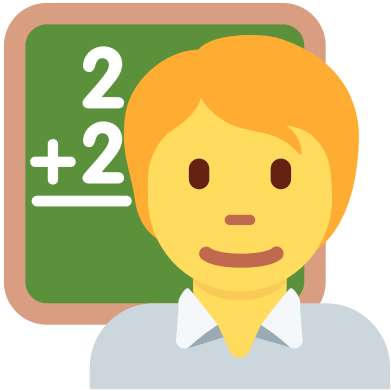
while loops



Code Detective

Find the bug 

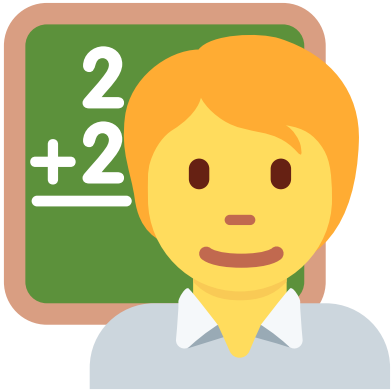
```
i = 0
while i < 5:
    i += 1
    print(i)
    if i % 2 == 0:
        continue
```

Human Calculator

What is printed?

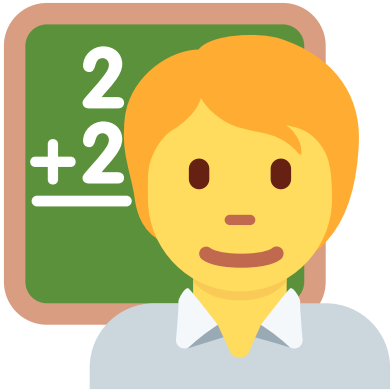
```
i = 3
while i > 0:
    print(i, end="")
    i -= 1
print("done")
```



Human Calculator

How many times does the loop run?

```
n = 0
while True:
    n += 1
    if n == 3:
        continue
    print(n)
```

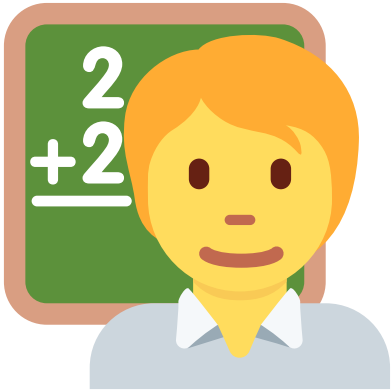


Human Calculator

What is printed?

```
i, s = 0, 0
while i < 5:
    i += 1
    if i % 2 == 0:
        continue
    s += i
print(s)
```

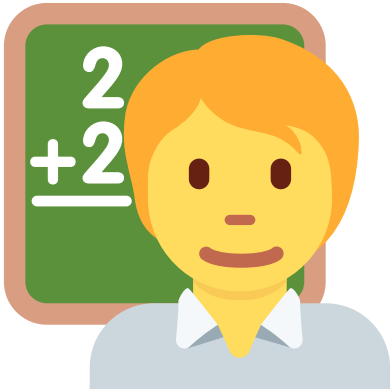
for loops



Human Calculator

What is printed?

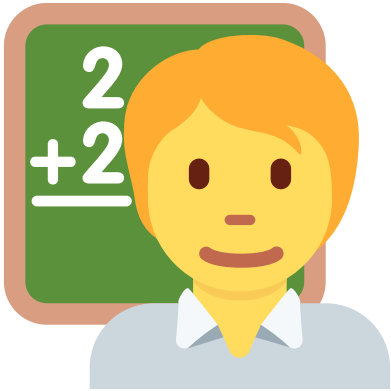
```
names = ["A","B"]  
for i, n in enumerate(names, start=1):  
    print(i, n)
```



Human Calculator

What is printed?

```
for x in [0,1,2]:  
    if x:  
        continue  
    print(x)
```



Human Calculator

What is printed?

```
for i in range(5):  
    print(2**i)
```



Code Detective

Find the bug 

```
# print odd numbers from 1 to 5 [1, 3, 5]  
for i in range(5, 1, 1):  
    print(i)
```